

ASSIGNMENT 2 – EXPLORE & BUILD AN N8N TEMPLATE

AI-Powered Customer Feedback Sentiment Analysis & Escalation

Selected Template Use Case: Automating real-time customer feedback categorization, AI sentiment analysis, urgent Slack escalation for negative reviews, and structured Google Sheets logging.

1. Template Overview & Business Problem

- **Problem Statement:** Customer Experience (CX) teams receive hundreds of feedback submissions daily via web forms and app surveys. Manually reading each submission to identify dissatisfied customers causes response delays, resulting in higher churn.
- **Solution Architecture:** An automated n8n pipeline that instantly ingests feedback via Webhook, standardizes payload data, evaluates sentiment using OpenAI (GPT-4o), conditionally triggers an immediate Slack alert to the support lead if negative, and archives all data into Google Sheets for reporting.

2. Node-by-Node Technical Breakdown

Node Name	Node Type	Role & Purpose in Workflow
1. Webhook	n8n-nodes-base.webhook	Acts as the trigger node. Exposes an HTTP POST endpoint receiving raw JSON payloads from frontend feedback forms.
2. Edit Fields (Set)	n8n-nodes-base.set	Data normalization node. Extracts <code>customer_id</code> , <code>email</code> , and <code>feedback_text</code> while adding a ISO timestamp.
3. OpenAI Agent	@n8n/langchain.openAi	AI Processing node. Analyzes <code>feedback_text</code> and outputs JSON with classification (<code>POSITIVE</code> , <code>NEUTRAL</code> , <code>NEGATIVE</code>) and a 1-sentence summary.
4. If (Conditional)	n8n-nodes-base.if	Logic routing node. Checks if sentiment equals <code>NEGATIVE</code> . True branch routes to Slack; False branch routes to logging.
5. Slack Alert	n8n-nodes-base.slack	Escalation node (True Branch). Posts formatted alert with customer details and AI summary to <code>#customer-escalations</code> channel.
6. Google Sheets	n8n-nodes-base.googleSheets	Database/Storage node (False Branch / Fallback). Appends sentiment tags, customer ID, and text to a master Google Sheet.

3. End-to-End Workflow Execution Steps

- **Step 1 (Ingestion):** Webhook receives payload: `{"customer_id": "CUST_902", "email": "user@example.com", "feedback_text": "App crashed twice during checkout!"}`
- **Step 2 (Data Prep):** Edit Fields node sanitizes parameters and injects server timestamp.
- **Step 3 (AI Inference):** OpenAI evaluates text $\rightarrow$ outputs: `{"sentiment": "NEGATIVE", "summary": "User experienced multiple app crashes at checkout."}`
- **Step 4 (Routing):** If node detects `NEGATIVE` sentiment and evaluates path as **TRUE**.
- **Step 5 (Action):** Slack bot publishes urgent escalation card to `#customer-escalations` for immediate team outreach.

4. How to Import the Recreated Workflow into n8n

- **1. Download Template File:** Use the generated .json workflow template file accompanying this submission.
- **2. Open n8n Canvas:** Log into your n8n instance and open a fresh workflow canvas.
- **3. Import via Menu:** Click on **Workflow Options (top right)** $\rightarrow$ **Import from File** and select `n8n_Customer_Feedback_Workflow_Template.json`.
- **4. Configure Credentials:** Attach your OpenAI API Key, Slack Bot Token, and Google OAuth credentials to their respective nodes.
- **5. Activate Workflow:** Toggle workflow status from Inactive to Active.